SYDNEY BEACHWATCH - WATER QUALITY & RAINFALL ANALYTICS (1991–2025)

A data-analytics portfolio project using R (cleaning + EDA), and Tableau (interactive dashboard)

Data: *NSW Beachwatch (water quality) and daily Sydney weather observations, 1991–2025*.

Live dashboard: Sydney Beaches Water Quality and Weather Records on Tableau Public

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EXECUTIVE SUMMARY

This project analyses **34 years of water-quality monitoring (1991–2025)** across **79 Sydney swim sites**, joined to a matching daily weather record, to answer a simple public question: how clean is the water at Sydney's beaches, and what makes it worse?

From **123,510 water samples**, **84.56%** met the safe-swimming guideline, **10.76%** were elevated, and **4.68%** breached the pollution-alert level. The headline safety figure is reassuring, but it hides substantial variation between sites, seasons, and, most importantly, **rainfall conditions** which are the dominant driver of contamination events. The work is delivered as a reproducible R pipeline and a published, interactive Tableau dashboard.

BACKGROUND

NSW's **Beachwatch** program monitors recreational water quality using enterococci - a faecal-indicator bacterium reported as colony-forming units per 100 mL (cfu/100 mL). Elevated enterococci signals possible sewage or stormwater contamination and raised illness risk for swimmers. Following established NSW/NHMRC guidance, each sample is classified into a traffic-light band:

BAND	ENTEROCOCCI LEVEL	MEANING
🟢 Good	<= 40	Suitable for swimming
🟡 Fair	<= 200	Swim with caution
🔴 Poor	> 200	Not safe for swimming

The project pairs this water data with daily weather (temperature and rainfall) because contamination in Sydney is strongly rain-driven: stormwater flushes pollutants into waterways, and bacterial levels spike in the following one to two days.

DATA

DATASET	RECORDS	TIME PERIOD
water_quality.csv	123,510 (after deduplication)	Sep 1991 – Apr 2025
weather.csv	12,538	Jan 1991 – Apr 2025

Water-quality fields: region, council, swim_site, date, time, enterococci_cfu_100ml, water_temperature_c, conductivity_ms_cm, latitude, longitude.

Weather fields: date, max_temp_C, min_temp_C, precipitation_mm, latitude, longitude.

DATA CLEANING & PREPARATION (R)

The two files sat at opposite ends of the quality spectrum, and the cleaning strategy reflected that.

a) Water quality

Exploring the dataset exposed several issues, each addressed explicitly rather than with a blanket fix:

- **Typing / sensor errors in temperature.** 26 readings fell outside a plausible Sydney range (up to 1040 °C). These were nulled before any statistic was computed, so they could not distort the imputation logic.
- **Conductivity anomalies.** A zero reading and a handful of impossible highs were nulled. The column is labelled mS/cm but its values are really μS/cm (seawater ≈ 53,000) - documented as a units caveat rather than silently rescaled.
- **Extreme enterococci kept.** The largest bacterial readings are genuine post-rain pollution spikes -signal, not noise - so they were retained.
- **20 exact duplicate rows** removed; text fields trimmed; dates and times parsed to proper types.

b) Missing-value strategy

Missingness was very uneven: enterococci was only **0.25%** missing, but water temperature (~61%) and conductivity (~64%) were mostly absent. A single skewness rule was applied: if skewness > 1, impute the median; otherwise the mean. Crucially, this rule was applied to **enterococci only** (heavily right-skewed → median). Temperature and conductivity were **left as NA**. Imputing 60%+ of a column with one constant would have flattened its variance and manufactured tens of thousands of identical fake readings, which is indefensible in any review. Those two fields are therefore treated as *contextual* only; every headline result rests on enterococci.

c) Weather-near-pristine

The weather series required almost no repair: no missing values, no duplicate dates, no calendar gaps across 34 years, and all readings physically plausible (max never below min). The same disciplined pipeline was run for consistency and reported zero fills.

d) Feature engineering

Water quality gained the traffic-light water_quality category, an is_safe flag, and calendar parts (year, month, season). Weather gained derived temperatures, hot-day flags, rainfall bands, and the analytical bridge between the datasets was **antecedent-rainfall features**: previous-day rain (rain_lag1, rain_lag2) and a rolling three-day total (rain_3day), the columns that let the rain-versus-pollution story be told.

EXPLORATORY FINDINGS

Enterococci is extreme-valued and right-skewed. Most samples are low, but a long tail of high readings drives the pollution story, which is why the median, not the mean, anchors the analysis.

The 84.56% safe rate masks site-level spread. Aggregate safety is high, but the riskiest sites, largely enclosed estuarine and harbour locations with poor flushing, carry a much higher share of "Poor" samples than open surf beaches. The dashboard's ranking panel surfaces the 10 worst offenders.

Contamination is seasonal and rain-driven. Exceedances cluster in the wetter months, and the rain-versus-pollution panel shows the exceedance rate rising steadily across antecedent-rainfall bands, dry days sit near the baseline, while sites following heavy multi-day rain show markedly higher breach rates. This is the project's central, actionable insight: rainfall is the strongest predictor of unsafe water.

A long-run time trend is visible in the yearly-exceedance line, letting the viewer judge whether water quality has improved, worsened, or held steady across three decades.

Exact per-year, per-season, and per-rainfall-band percentages are read directly from the live dashboard; the patterns above describe their direction and are stable under filtering.

DASHBOARD DESIGN (TABLEAU)

The dashboard is built on a **relationship** between the two clean tables joined on date (many samples → one weather day), which keeps each table at its native grain and avoids double-counting. Key logic is implemented as Tableau calculated fields and LOD expressions, such as a {FIXED [Swim Site]} site-median classifier that colours each map marker, and ratio measures (% Safe, % Exceedance, % Poor) that are safe to average across the join.

Layout (fixed 1200×800): a title and global filter strip (Region, Site Status, Year), a four-KPI row, a Beachwatch-style map as the hero (green/amber/red markers, sized by sample count, acting as a filter on click), a top-10 riskiest-beaches ranking, and a bottom row of three analytical charts -yearly trend, seasonal pattern, and rain vs pollution. Filters apply across the whole workbook and cascade via relevant values.

LIMITATIONS

- **Single weather station.** One central-Sydney gauge is applied to all sites; localised rainfall varies across the metropolitan coast, so the rainfall link is directional rather than site-exact.
- **Sparse secondary variables.** Temperature and conductivity are ~60% missing and left un-imputed; they support context, not conclusions.
- **Per-sample classification.** The traffic-light bands are applied per sample for interpretability. Official Beachwatch beach grades use seasonal percentile statistics, which is a stricter, separate calculation.
- **Sampling is not daily per site**, so site time series are irregular.

CONCLUSION

Sydney's beaches are safe to swim the large majority of the time, but safety is conditional -it drops around specific sites, seasons, and especially after rain. The project demonstrates an end-to-end analytics workflow: rigorous, decision-driven data cleaning in R; honest handling of missing data; feature engineering that connects two datasets; and a published, interactive Tableau dashboard that turns 123,510 rows into a public-facing tool a swimmer could actually use.